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Understanding consumer engagement in social media: The role of product lifecycle

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ABSTRACT

More than 4 billion unique users are now using different social media platforms. This provides firms with a new avenue to connect and converse with their targeted customers in a dialogue fashion and create better customer engagement. Although customer engagement has been studied, no study to date has focused on the concept of customer engagement in social media through the lens of users' participation in (co)creation of content on social media. To that end, this study strives to conceptualize and validate the concept of customer engagement with a social media post, proposing a research model which investigates the effects of popularity, discourse logic and argument frame on customer engagement with a social media post. This study also investigates to what extent those effects vary by the product lifecycle stage and social media platform type, using secondary data extracted from Instagram and Twitter on a group of products (i.e., vehicles) at different lifecycle stages to address its research objective. This study's findings would help firms make better decisions on enhancing their customer engagement on social media platforms.

1. Introduction

Social media platforms enable their users to build their virtual communities while sharing their ideas or thoughts [1]. These abilities have motivated billions of individuals across the globe to join at least one of the social media platforms. Recent reports estimate that by January 2021, the total number of unique users across various social media platforms has reached 4.2 billion, which shows a 13% increase compared to the previous year (i.e., 2020) [2]. This report also states that each social media user, on average, spends 2 h and 25 min per day browsing her/his social media page(s). Moreover, Tankovska [3] notes that, in 2020, the daily number of shared stories on Instagram has reached 500 million, almost three times the number of stories shared in 2018.

Social media use allows a firm to control its information broadcast, connect and converse with its targeted customers in a dialogue fashion and create better customer engagement [4]. According to a study by Buffer, 2019, the significant majority of marketers (i.e., 73%), who participated in that study, believed that social media marketing is the most effective means for firms to promote their products [5]. Steven Brill, CEO of NewsGuard, says that "Social media moves and shapes the public dialogue and companies operating within it are responding more to their impact on society" [6]. A firm's presence in social media could potentially strengthen its engagement with customers [7]. A higher level of customer engagement allows a firm to increase its website traffic, sales, customer satisfaction, loyalty, and consequently decision quality [8–10].

Various groups of scholars and practitioners have vastly studied the concept of consumer engagement and also pointed out guidelines that a firm could put into practice to increase its customers' engagement [11]. However, there is a continued dialogue among scholars and practitioners of Information Systems and Marketing on the accuracy, validity and reliability of the means used to measure consumer engagement, especially in the context of social media [9,12]. Conventionally, consumer engagement entails customers cognitive and physical connections with a brand [13]. In the online environment, firms use interactive approaches to increase their customers' physical experiences with their products/services. As opposed to physical experience enhancement, customer engagement through social media allows a firm to impact their customers' emotional and cognitive nuance through motivational drivers such as customer reviews or social media posts [13]. In this regard, customer engagement refers to making customers voluntary have

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contributions to a brand's image, beyond any physical transactions [14]. Recent studies confirm that customers engaged with a brand on social media are three times more likely to remain loyal to a brand [13]. In the context of this study, *customer engagement* reflects a customer's willingness to participate in the dialogue related to a specific product [11]. While increasing consumer engagement is critical for firms, there is still a lack of understanding of how to conceptualize, validate or measure consumer engagement in social media in the context of users' participation in content (co)creation for a product. Therefore, as the *first objective*, this study strives to define, conceptualize, measure, and validate the concept of customer engagement in social media in the context of users' (co)creation of content related to a product, as a proxy to measure the actual customer engagement with that specific product.

Traditionally, firms have employed various methods to enhance customer engagement [15], among which is the use of renowned public figures known as influencers to promote their products [16]. According to Source Credibility Theory [17], a promoting message's credibility depends on the receivers' perception of the influencer's trustworthiness and reliability. Renowned influencers are popular and have many fans who follow them, especially on their social media pages. Extant literature purports that an influencer's popularity enhances the credibility of her/his promotion, positively impacting the potential customers' decision to acquire a particular product (e.g., [18]). Moreover, extant literature also mentions that popularity is a proxy to classify social media users as influencers versus ordinary users [19]. That said, this study considers the number of followers of a user in social media as a proxy to portray her/his *popularity* (i.e., being an influencer or an actual user).

Extant literature also finds a promoting message's argument quality as another factor shaping and enhancing customer engagement [20]. According to the Source Credibility Theory [17], a promoting message's argument quality is evaluated by its trustworthiness and attractiveness. This study proposes that the argument quality of a message in social media depends on two primary attributes of *discourse logic*, reflecting trustworthiness [21], and *argument frame*, representing attractiveness [22,23]. Discourse logic shows whether an influencer, in her/his message, endorse a product providing objective evidence instead of her/his subjective opinion. Argument frame reflects positivity, negativity, or neutrality of the tone of the influencer's message/recommendation regarding a product. Hence, this study's *second objective* is to inspect the extent to which *popularity*, *discourse logic*, and *argument frame* affect customer engagement in social media.

This study argues that the importance of popularity, discourse logic, and argument frame on consumer engagement may depend on products' *lifecycle* stages (i.e., introduction, growth, maturity, and decline) [24]. According to Armstrong et al. [25], the type of media, promotion and messages effectively endorsing a product varies by its lifecycle stage. Therefore, as the *third objective*, this study explores how a product's lifecycle stage moderates the proposed relationships outlined in our second objective.

Furthermore, users' demographics vary among social media platforms [26]. For instance, most users on Instagram (i.e., 80%) are less than 30 years old, compared to Twitter, in which most users (i.e., 60%) are above 30 years old [22]. According to Leone [27] customer engagement varies by platform. For instance, among the social media platforms, Instagram has the highest level of engaged social media users (i.e., 7%) interacting with a social media post, while Twitter has the least engaged users (i.e., less than 2%) with a social media post promoting a product. Therefore, customer engagement and its antecedents could vary by social media platform, and, thus, in the post hoc analysis, we examine whether the relationships shown in the research model vary for Instagram and Twitter as the *fourth objective*. This study will specifically focus on social media posts on Instagram and Twitter since they are among the most used social media platforms by firms to promote their products [22].

This study relies on Source Credibility Theory [17] and the Systematic/Heuristic Persuasion Model [28] to explore the above research objectives. We particularly utilized these theories because they enable us to understand whether popularity, which represents a promoter's credibility, discourse logic that is systematically evaluated, and argument frame, which is heuristically evaluated, affects social media engagement. Secondary data containing promotive messages for products in different lifecycle stages is scrapped, extracted, and analyzed from Instagram and Twitter to address the above research objectives.

2. Literature review and theoretical background

2.1. Customer engagement

Traditionally, scholars and practitioners of Marketing have studied the concept of customer engagement, its definition, barriers, drivers and antecedents [11,29,30]. Particularly, Marketing literature suggests "customers' interactive and co-creative actions entails customer engagement" [13]. Kumar and Pansari [31] define customer engagement as the level of connectedness between a customer and a firm, stating that a higher level of customer engagement enhances a firm's competitiveness. Vivek et al. [32] expand the above definition, asserting that customer engagement also involves "events and activities engaged in by the customer that are not directly related to search, alternative evaluation and decision making involving brand choice". According to Roberts and Alpert [29], the main driver of enhanced customer engagement is the alignment between a firm's value proposition and its customer experience activities. Roberts and Alpert [26] also define five different stages an engaged customer goes through to connect with a product, with each of these stages building on the previous one. These stages are: product purchase transaction, continued repurchases, readiness to purchase other lines of products, recommending to others, and finally becoming a brand's advocate. To summarize, customer engagement in the marketing literature lingers around a firm's ability to connect with its customers. It is worth noting that the above stages differ from those in a product lifecycle. A customer goes through the above phases of engagement with a product at any product lifecycle stage.

The widespread use of social media introduced new opportunities for firms to connect with their customers directly. It also attracted various information system scholars' attention to study the concept of customer engagement in social media [33–35]. According to Cabiddu et al. [36], firms can plan three types of customer engagement strategies: persistent, customized, or triggered. The persistent customer engagement strategy outlines the plans for continued discussions with the customers. The customized customer engagement strategy portrays the goals for a tailored customer interaction based on prior knowledge of customers' interests, preferences, conversations, and relationships. The triggered customer engagement strategy is applicable when a firm plans for customer-initiated events to explore and identify new value-creation opportunities. All in all, engaging with customers in social media enables a firm to have more loyal customers [37], to increase its performance, sales and referrals [38], and to improve its brand awareness [39].

Extant literature traditionally offers various methods to conceptualize and measure customer engagement [40], mainly grounded in surveys focusing on how physically customers connect and engage with a product. With the emergence of new technologies, such as recommender systems, online consumer reviews and social media platforms, the focus of customer engagement studies leaned towards behavioral interactions of customers with a product [13]. Having said that, the definition of customer engagement also changed to "customers making voluntary resource contributions that have a brand or firm focus but go beyond what is fundamental to transactions" [14]. While the definition has been changed, the methods to measure customer engagement in social media remained the same, relying on surveys [40] which is a proxy to customer engagement. However, based on the above definition, customer engagement can be measured by how customers react to a

social media post. Therefore, there is still a lack of understanding of how to conceptualize, validate, and measure the concept of customer engagement in social media through the lens of customers participation in content (co)creation. According to the extant literature, common factors on social media platforms reflecting engagement with a post include the number of likes, comments, and shares [41]. In that sense, this study proposes, conceptualizes, validates, and measures the concept of customer engagement in social media as a multidimensional formative construct. Our proposed customer engagement construct comprises the total number of *likes*, *comments*, and *shares* of a social media post. The new offered method in this study allows a firm to directly measure to which extent their current or future potential customers are responding or will respond towards their marketing efforts. It also enables a firm to update its marketing strategies and plans accordingly in a timelier fashion, resulting in a higher customer engagement level since customers can perceive the changes as they occur just in time.

2.2. Source credibility theory

Source Credibility Theory [17] offers a theoretical explanation about how a message is assessed for its credibility. According to this theory, the receiver evaluates both the messenger and the message's credibility, attractiveness, trustworthiness, and quality. According to this theory, a receiver first estimates the messenger based on her/his attractiveness, reliability and trustworthiness [42]. According to McCroskey and Jenson [43], a credible messenger is an individual with the following five characteristics: competence, character, sociability, composure, and extroversion. Competence refers to the qualification and expertise of a messenger in the domain area of the message. Character portrays a messenger's sympathy and selflessness. Sociability illustrates a messenger's temperament and friendliness. Composure shows a messenger's excitability. Finally, extroversion refers to the extent to which a messenger is frank, bold and fair, debating a resolution [39]. In the general public's view, renowned public figures who have these characteristics are deemed to be more trustworthy, attractive and credible [44]. One way to measure credibility in social media is by the popularity of a social media user [45].

A receiver also evaluates the message's quality based on two factors of discourse logic and argument frame [46]. Discourse logic shows whether a social media user has provided objective evidence endorsing a product instead of subjective opinion on her/his social media post, reflecting the trustworthiness of a post. Argument frame illustrates the user's tone reflecting her/ his positive, negative or neutral opinion about a product offered in a social media post, portraying the attractiveness of a post.

2.3. The heuristic / systematic model of persuasion

The Heuristic / Systematic Model of Persuasion (HSM) [28] explains to what extent a receiver of a message processes it. According to the HSM, there are two different information processing streams: systematic and heuristic information processing [47]. Systematic processing is used when an individual thoroughly processes all the available information and cues in an argument (e.g., the reasoning behind the statement or the person who is arguing), using deep thinking and intensive reasoning. As opposed to the systematic processing of information, heuristic information processing requires less deliberative information processing. Heuristic processing of information is generally automatically used by an individual. When heuristic processing is applied, an individual only pays attention to noticeable cues of information in an argument, such as an arguer's characteristics, the number of reasons presented in a statement, or positive or adverse reactions of an arguer [43]. The heuristic/ systematic model of persuasion has been vastly used by various Information Systems scholars [48,49]. For instance, Wathen and Burkell [50] have used HSM to identify the factors that affect website credibility. Their study states that assessing a website's credibility starts with high-level processing of information. At first, individuals examine the peripheral cues such as the appearance, design, or reputation of a website. Then, they go through a thorough low-level analysis of information. In the end, the credibility results in an interaction of the outcomes of both high and low-level information processing (i.e., heuristic and systematic processing of information)[1] [46].

HSM and Source Credibility Theory are two theories that go hand in hand. They are mainly used together to analyze the credibility, trustworthiness and attractiveness of presented information. This study assumes that the heuristic processing of information is used when an individual responds to a social media post's argument frame or an influencer's popularity because the positive or negative tone or the number of followers can be evaluated quickly by a social media user. On the other hand, the systematic processing of information is used on occasions when a social media user thoroughly looks for objective evidence in a post (i.e., objective discourse logic). Therefore, the user should spend a significant amount of time validating the influencer's facts in the post.

3. Research model and hypotheses development

Fig. 1 depicts the research model of this study which shows the impact of popularity, discourse logic, and argument frame on customer engagement in social media while considering the moderating role of the product lifecycle.

Table 1 provides an overview of the theories used in this study and their association with the research model's proposed hypotheses.

The Popularity of an individual in social media is measured by the number of her/ his followers [51]. Popular renowned figures are referred to as social media influencers. By definition, a social media influencer is "a user who has established credibility in a specific industry, has access to a huge audience and can persuade others to act based on their recommendations" [52]. Social media Influencer could be a blogger on Twitter or a public known celebrity on Instagram, followed by many social media users. Social media Influencers have the authenticity to attract many users' attention, gaining a higher reachability level [52]. Due to influencers' higher level of reachability, it is expected that a social media post by these individuals will draw the attention of a larger group of social media users (i.e., a higher level of views) [53]. Extant literature finds a positive correlation between social media post's views and social media users' reactions to such a post [54]. It is deemed that a higher level of view attracts more likes, comments and shares by other social media users [55]. Hence, a post by a social media influencer is expected to attract more comments, likes, and shares (i.e., a higher customer engagement level).

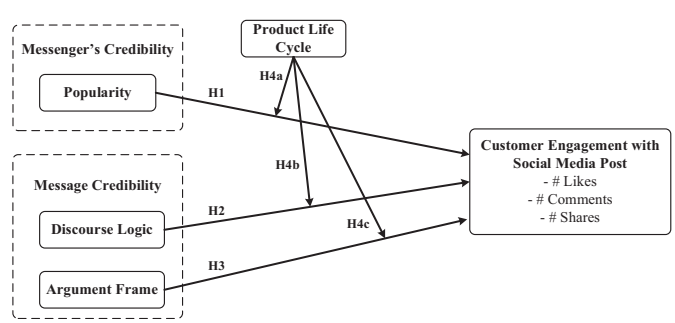

Fig. 1. Research model.

[1] HSM is very similar to the Elaboration Likelihood Model (ELM), except that ELM also explains two main information processing routes: central and peripheral.

Table 1
Hypotheses and theories association.

Hypothesis	Theory used to support	Customers interests to engage
H1	Source Credibility Theory HSM- Heuristic Information Processing	Messenger's Credibility Less Extensive Information Processing
H2	HSM – Systematic Information Processing	Message's Trustworthiness Extensive Information Processing
H3	HSM- Heuristic Information Processing	Message's Attractiveness Less Extensive Information Processing
H4a	HSM- both Systematic and Heuristic Information Processing	Introduction Product Lifecycle Stage Increase familiarity Provide product details
H4b	HSM- Systematic Information Processing	Growth and Maturity Product Lifecycle Stage Message's Trustworthiness Extensive Information Processing
H4c	HSM- Heuristic Information Processing	Decline Product Lifecycle Stage Message's Attractiveness Less Extensive Information Processing

Extant literature also shows that a higher level of customer engagement leads to more brand awareness, recommendations, referrals, sales, and profit [37,56,57]. Influencers, in general, and in the context of this study, social media influencers have been traditionally used by firms to increase customer engagement [58]. According to Source Credibility Theory [17], social media influencers are credible, trustworthy and attractive [59]. Moreover, according to HSM [28], popularity is considered a peripheral cue, using heuristic information processing, requiring less extensive information processing, to increase attention on a social media post. Popularity is also a proxy to measure a promoter's credibility and provides higher reachability of the message to a larger audience [55]. Hence, if a social media influencer, with many social media followers, provides a social media post about a product, a higher number of social media users is expected to react to that post. A higher level of reaction brings more likes, shares, and comments, leading to a higher level of customer engagement for the product posted. Hence, it is hypothesized that:

H1. *: A higher level of popularity for a message poster is positively associated with customer engagement in social media.*

Discourse Logic entails whether a social media post promoting or disapproving a product contains objective evidence versus subjective opinion about a particular product [60,61]. According to Source Credibility Theory [14], to deem a message (e.g., a social media post) credible, an individual (e.g., a social media user) assesses it based on the two merits of trustworthiness and attractiveness. A social media post's discourse logic represents the trustworthiness attribute, while a social media post's argument frame represents the attractiveness attribute of a message's credibility, respectively.

According to Source Credibility Theory, a message is deemed to be trustworthy if the messenger supports her/his statement by objective and tangible evidence(s). However, evaluating whether a statement is objectively supported is timely. According to HSM [28], to assess central cues of information in a statement, a receiver spends more time evaluating it using systematic information processing. Therefore, a social media user would spend a significant amount of time assessing whether the social media post is trustworthy (i.e., supported by tangible evidence (s) instead of holding the subjective opinion of the messenger). If the social media user finds a post trustworthy, she/he is more likely to follow its recommendations [62]. According to Coleman [63], trustworthiness, as the primary driver of customer engagement [64], endorses a higher level of social exchange discussions, leading to a higher customer engagement level. Therefore, an objectively supported social media post is expected to increase customer engagement, attracting more likes, shares, and comments. Hence, it is hypothesized that:

H2. *: A higher level of objectivity of a social media post's discourse logic is positively associated with customer engagement in social media.*

Argument Frame expresses the tone (i.e., positivity, negativity or neutrality) of a post on social media, reflecting the attractiveness attribute of a credible message [22]. As a peripheral cue, heuristically measured, extant literature finds positively written social media posts to be more attractive compared to the neutrally or negatively written ones, drawing the attention of more social media users to participate in social exchange discussions [65]. Moreover, positively written social media posts encourage other social media users to interact with a brand, creating a broader social media community and higher customer engagement in social media [66]. Additionally, customers perceive positively written social media posts promoting a brand to be more attractive while finding them more authoritative than negatively or neutrally written social media posts. According to Source Credibility Theory [14], a higher level of attractiveness leads to a higher perception of the message's quality and credibility. As stated earlier, a message's credibility (e.g., in a social media post) is the primary driver of customer engagement. Hence, it is expected that the positively (i.e., promotive) written social media post brings more customer engagement on a product in social media, attracting more likes, shares, and comments, comparing to a neutrally or negatively written one. Hence, we hypothesize that:

H3. *: A higher level of positivity of a social media post's argument frame is positively associated with customer engagement in social media.*

A product lives through different periods, referred to as **Product Lifecycle** stages. These stages are classified as *introduction*, *growth*, *maturity* and *decline* [24]. To effectively endorse a product, the choice of media, promotion, and messages depends highly on the products' lifecycle stage [25]. A product that has been newly introduced to the market has a low demand [67], given customers less familiarity with it [68]. Therefore, the marketing objective is to disseminate information about the new product and to present it to the general public as much as possible [69]. One way to increase customer engagement for a product at an introduction phase is to select a brand ambassador whose voice would be seen/heard among a broader community [70]. The primary role of the brand ambassador is to increase the social media users' awareness about a product's attributes [71]. The entire idea of choosing an ambassador is to increase brand awareness and encourage other social media users to get passionate about engaging and promoting such a product on their social media platforms [70]. Therefore, the introduction stage's best marketing strategy is to hire renowned public figures to promote and disseminate information about the newly introduced product since they have a higher number of followers and a higher potential to increase a product's market reachability [72]. Accordingly, it is expected that at the introduction stage, customer engagement in social media would be higher for those social media posts shared by famous renowned figures compared to products in later lifecycle stages. Hence it is hypothesized that:

H4a. *: Product lifecycle moderates the positive relationships between popularity and customer engagement in social media such that the impact of popularity on customer engagement increases when a product is in the introduction stage of its lifecycle compared to other stages.*

Customers at the growth and maturity stages presumably have become significantly familiar with the product, and it is presumed that the demand for the product should have risen to its highest level [73]. In these two stages of the product lifecycle, customers will pay attention to the details provided and spend much time reading about the product details [74]. Based on HSM, customers systematically process information about a product to evaluate its quality and credibility at the growth

and maturity stages. Therefore, it is anticipated that a social media post that is objectively supported brings the attention of many social media users who are likely to acquire the product. Accordingly, it is expected that customer engagement in social media would be higher for those posts supported by tangible objective evidence(s) at the growth and maturity product lifecycle stages compared to products in other lifecycle stages. Hence, it is hypothesized that:

H4b. *: Product Lifecycle moderates the positive relationship between discourse logic and customer engagement in social media such that the impact of discourse logic on customer engagement increases when a product is at the growth or maturity stage of its lifecycle compared to other stages.*

A product would be at its decline stage when demand starts going down. This stage lasts until the product is completely retired from the market [67]. At this stage, firms try to send positive promotive messages to customers to gain their attention back to the product and convince and attract them to acquire it [67]. Therefore, positively promotive messages (e.g., positively written social media posts) are more likely to attract potential customers (e.g., social media users) attention [75]. Therefore, exposure to positively promoted social media posts is presumed to attract a significantly higher number of social media users' attention when a product is at its decline stage of the product lifecycle [21]. Accordingly, customer engagement in social media would likely be higher for those posts that are supported by positively-framed arguments at the decline product lifecycle stage compared to any of the other stages. Hence, it is hypothesized that:

H4c. *: Product Lifecycle moderates the relationships between argument frame and customer engagement in social media such that the impact of argument frame on customer engagement increases when a product is at the decline stage of its lifecycle compared to other stages.*

4. Research methodology

4.1. Data collection

This study uses secondary data, scraped using web-scraping techniques from two social media platforms of Instagram and Twitter to test the research model and its associated hypotheses. The scrapped unprocessed data included posts' textual content, which was subsequently used to extract discourse logic and argument frame. The number of likes, comments and shares of a post, as formative indicators, was scraped to conceptualize and measure customer engagement in social media. Moreover, the number of followers of those whose posts were scraped, indicating popularity. The general theme selected was social media posts regarding vehicles. The main reason for this choice of product was that there are precisely distinguishable vehicles at various lifecycles stages. In this regard, to test the moderating effect of the product lifecycle, four groups of vehicles were identified: (1) self-driving (i.e., introduction stage), (2) full electric (i.e., growth stage), (3) hybrid cars (i.e., maturity stage), and (4) gasoline/diesel cars (i.e., decline stage) [76]. The introduction stage is when a product endeavours to get noticeable attention through promotions or words of mouth on social media [69]. An example of a product at its introduction lifecycle stage is self-driving cars. At the growth product lifecycle stage, the product's visibility increases, mainly through similar introduction stages marketing incentives. An example of a group of vehicles at the growth product lifecycle stage is full electric cars [69]. The maturity product lifecycle stage is when a product reaches its highest profitability and attention. Mainly at this stage, manufacturers roll out new features to amend their customers' interests. An example of this group of products is hybrid cars [69]. A product at its decline stage has lost its profitability. Marketers mainly prepare a sentimental positive marketing campaign to delay the farewell of a product from the market. An example of a vehicle at this stage is gasoline/diesel cars [69].

According to the Uses and Gratifications Theory (UGT) [77], people use different social media platforms to satisfy different needs. For instance, according to [78], people choose Instagram to socialize, entertain, or seek information about a particular product. On the other hand, Twitter is primarily used for sharing ideas [26]. Accordingly, to satisfy this study's fourth objective, data was scraped from both Instagram and Twitter. These two social media platforms were chosen because various firms primarily use them to increase customer engagement in social media [79,80].

Moreover, the type of social media users between these two platforms varies a lot. While on Twitter, the majority of users are older than 30 years old (i.e., 60%), on Instagram, the majority of users are less aged less than 30 (i.e., 80%) [23]. This significant difference in users' demography on these two platforms implies that these two age groups have different needs and wants. Therefore, firms should adopt a significantly different approach to increase customers' engagement in social media between these two platforms [81]. Hence, in pursuit of our fourth objective, we would strive to understand any differences between the relationships shown in Fig. 1 for Instagram and Twitter.

Having said that, in this study, we collected social media posts that met the above criteria for a period of four years, from 2016 to 2020. Our final dataset yielded 3324 posts, where 1583 social media posts were extracted from Instagram, and 1741 social media posts were extracted from Twitter. Table 2, provided in Section 5.1, illustrates the summary of descriptive statistics of our collected dataset.

4.2. Text mining to extract discourse logic and argument frame

Following the extant literature's recommendations, two different classifiers were implemented to extract discourse logic (i.e., objectivity vs subjectivity) and Argument Frame (i.e., Positivity, Neutrality, or Negativity) of social media posts. This study followed the recommendations of Hajj et al. [82] on subjectivity analysis. Developed in Python, a classifier using a supervised support vector machine model, employing the parts of speech (POS) tagging technique, was implemented to determine whether the text of a social media post is objectively written or subjectively inscribed. The classifier was programmed to extract features related to subjectivity or objectivity written in a social media post [83]. The implemented code looks for quotations, past tense verbs, third-person pronouns, dates, and numbers to determine objectivity.

Table 2
Summary of descriptive statistics on both Instagram and Twitter extracted social media posts.

Construct	Mean	Median	Standard deviation	Min	Max
Likes	1574	136	20,782.93	0	1,100,000
Comments	76	7	285.41	1	7000
Shares	539	34	8724.20	0	431,000
Popularity	2,150,994	719,349	4,511,407.56	17	64,997,584
Discourse Logic	−0.29	−0.24	0.30	−1	1
Argument Frame	0.09	0	0.23	−1	1
Total Number of Social media Posts for Self-Driving Cars (Introduction Stage)					501
Total Number of Social media Posts for Full Electric Cars (Growth Stage)					957
Total Number of Social media Posts for Hybrid Cars (Maturity Stage)					775
Total Number of Social media Posts for Gasoline/Diesel Cars (Decline Stage)					981
Total Number of Social media Posts Extracted from Instagram					3324

Numbers associated with Likes, comments, shares represent the number of likes, number of comments, and shares on a post.
Popularity shows the number of followers of social media influencers.
Discourse logic: ranges between −1 to 1, where the positive number represents objectivity, and the negative number represents subjectivity.
Argument Frame: ranges between −1 to 1, representing negativity, neutrality, or positivity of the tone of a social media post.

Subjectivity is determined, detecting imperative and present tense verbs, comparative adjectives, first and second pronouns, and exclamation or question marks [75]. The implemented classifier contains a lexicon, a text processor, a trainer system, and a predictivity accuracy testing system [76,78], employing the outlined algorithm.

This study follows extant literature's recommendations [22,23,85–88] to extract the textual portion of social media posts' argument frame by a sentiment analysis technique to extract social media posts' polarity. There are different variations of sentiment analysis techniques. The choice of sentiment analysis technique varies the level of analysis and algorithm type [89]. Based on the level of analysis, sentiment analysis techniques are either concerned with determining the polarity of a given text (i.e., document-based sentiment analysis) or determining the polarity of a given text towards a specific feature (i.e., feature-based sentiment analysis). There are also three general groups of algorithms for sentiment analysis to extract the polarity of a given textual document: lexicon-based, rule-based, and machine learning [90]. Lexicon-based and rule-based algorithms use a lexicon to navigate the polarity extraction [19,84]. A machine-learning algorithm uses a deep learning approach to determine and classify a given text's polarity based on a set of criteria of interest [90-92]. This study used a machine-learning document-based sentiment analysis classifier implemented in R, using Sentiment Library, detecting the overall argument frame or polarity of the textual portion of a social media post. This classifier could classify a text, based on its polarity, into three categories: positively-written, neutrally-written, or negatively-written.

Two of the authors independently read and assigned a discourse logic, and an argument frame for a randomly chosen 200 social media posts sample to cross-validate the results. To assess the agreement between the two raters, we computed Cohen's kappa. Results showed inter-rater reliability of 0.91 for discourse logics and a 0.90 inter-rater reliability for argument frames, significantly higher than the accepted threshold of 0.6 [93]. Furthermore, to resolve discrepancies between the authors, two authors commutatively redid the above process to assign a discourse logic and argument frame for the chosen sample. Finally, we checked and compared the rated discourse logic and argument frames assigned by the raters versus the algorithms employed. The compassion results revealed a 93% accuracy rate for the discourse logic classifier and a 91% accuracy for the argument frame classifier. These two accuracy rates are significantly higher than the accepted threshold of 80% offered by the extant literature [22]. Hence, we concluded that both classifiers' outputs are reliable and can be used for further analyses. Fig. 2 depicts the research methodology employed in this study.

As shown in Fig. 2, this study took an explanatory approach relying on secondary data to explain how customer engagement in social media is shaped. Since this study aimed to understand how the textual component of social media posts brings customer engagement, the appropriate approach was to use a mixed-method approach, including quantitative and qualitative research methods. As explained earlier, the qualitative method was used to analyze social media texts and transform the textual document into quantitively measurable values to be used for further analyses. As suggested by the extant literature [94,95], the chosen textual analysis techniques matched this study's research objectives. In line with the explanatory research approach, the quantitative method was also used to explain to what extent the proposed hypotheses supported by the theories affect customer engagement with a social media post.

5. Data analyses and results

5.1. Summary of descriptive statistics

Accordingly, employing the above research methodology, Tables 2 depicts the summary of the descriptive statistics analysis results.

Tables 3 illustrates the correlations among the antecedents and indicators of customer engagement with a post on social media.

5.2. Validating customer engagement in social media as a formative construct

Hair et al. [96] suggested the two steps (i.e., indicator collinearity analysis and indicators statistical significance) should be conducted to examine a formative construct. As shown in Table 3, this study does not suffer from collinearity among the indicators of customer engagement with a social media post. As a final step, we used the bootstrapping method in Smart PLS 3.3.2 [97] to examine the significance of the first-order indicators (i.e., numbers of likes, comments and shares) in shaping customer engagement in social media. Tables 4 depicts the results.

According to Gefen and Straub [98], the outer model weights and T-values indicate the importance of indicators, forming customer engagement in social media as a formative construct. As shown in Table 4, the number of likes and shares are the most critical factors shaping the perception of customer engagement in social media. Surprisingly, it was also found that the number of comments does not significantly contribute to forming the concept of customer engagement in social media. Therefore, as recommended by the literature, we decided to drop it from the measurement model [83]. The analyses outlined above satisfy our first research objective to conceptually test and validate the concept of customer engagement in social media through the lens of social media users' participation in (co)creation of content.

We also checked for the discriminant validity results for the measurement model. Results are shown in Table 5. As shown in the Table, the results confirm that the collected data does not suffer from discriminant validity issues.

5.3. Structural model and hypotheses testing

Following the extant literature, this study employed the bootstrapping method with 1000 re-samples, using SmartPLS Version 3.3.2[2] to test the structural model and its associated hypotheses [99]. According to the extant literature, bootstrapping, a nonparametric method, allows testing the processed relationships' significance level in a structural model, assuming that the data is not normally distributed [96]. Fig. 3 illustrates the results of these analyses.

The analysis, shown in the above figures, revealed that a social media influencer's *popularity* and *objectively* written social media post is positively correlated with customer engagement in social media. Contrary to our initial hypothesis, we did not find any association between the argument frame and customer engagement in social media. Our analysis also reveals that, based on the effect size, an influencer's popularity is the most significant factor driving customer engagement in social media. In this regard, we used the f square value as the indicator of effect size. According to Cohen [100] guidelines, the f square value for the effect of influencer's popularity on customer engagement in social media showed a value of 0.288 (i.e., f square between 0.15 and 0.35), representing a moderate effect size. The f square value for discourse logic on customer engagement in social media was 0.024 representing a small effect size (i.e., less than 0.15). Moreover, a statement's objectivity of a social media post also increases customer engagement, while a message's tone does not improve customer engagement. Additionally, customer engagement's r squared values revealed that the research model meets the minimum accepted range for an acceptable r squared in social sciences studies (i.e., 10%) [101].

Finally, we tested our research model for its model fit. Our analysis yielded 0.018 for SRMR and 0.982 for NFI, which both satisfy the minimum accepted thresholds for a good model fit (i.e., SRMR < 0.08 and NFI > 0.95) [102]. Therefore, it could be concluded that the

[2] Ringle, Christian M., Wende, Sven, & Becker, Jan-Michael. (2015). SmartPLS 3. Bönningstedt: SmartPLS. Retrieved from http://www.smartpls.com

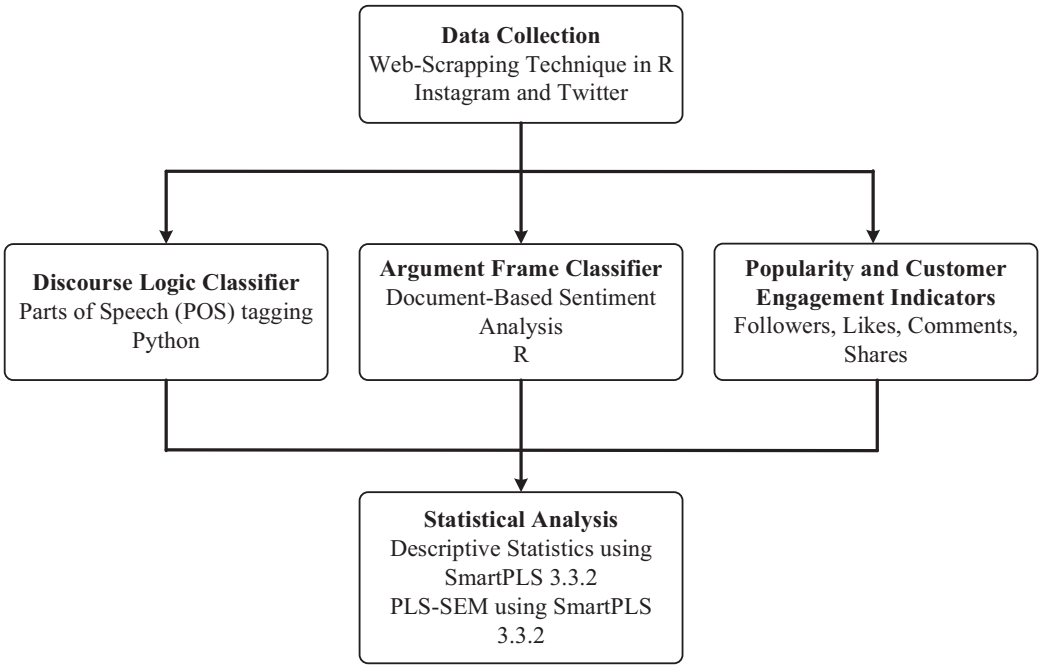

Fig. 2. Research methodology.

Table 3
Correlation matrix on social media posts extracted from both Instagram and Twitter.

	Like	Comment	Share	Popularity	Argument frame	Discourse logic
Like	1					
Comment	0.167	1				
Share	0.139	0.326	1			
Popularity	0.230	0.152	0.225	1		
Argument frame	0.0238	0.096	−0.0222	0.065	1	
Discourse logic	0.070	−0.051	0.073	0.015	−0.140	1

Table 4
Customer engagement in social media as a formative construct.

Relationship	Weight	T-value	Keep
Likes ➔ Customer engagement with a social media post	0.652	6.276	**Yes**
Comments ➔ Customer engagement with a social media post	0.114	0.888	**No**
Shares ➔ Customer engagement with a social media post	0.615	5.206	**Yes**

research model of this study does not suffer from the fitness of data.

5.4. The moderating effect of product lifecycle on the structural model's hypotheses

In line with our third objective, we checked to what extent product lifecycle moderates the research model's relationships. To that end, this study categorized the dataset in hand based on the product lifecycle stages and used the bootstrapping method with 1000 re-samples, using SmartPLS Version 3.3.2 to figure out whether the effect of customer engagement antecedents varies by product lifecycle stage. Results are shown in Table 6.

The above results confirm that, as hypothesized in H4a, if a product is in its introduction stage, product lifecycle moderates relationships between popularity and customer engagement in social media, increasing customer engagement in social media. Interestingly, it was found that product lifecycle also moderates the relationship between discourse logic and customer engagement in social media. It is deemed that customers are more interested in hearing objective evidence about a product's quality at the introduction stage. Therefore, they tend to pay more attention to a message's details [74]. According, as shown in Table 6, customer engagement in social media would be higher for those social media posts objectively supported and endorsed by famous renowned figures at the introduction stage.

As hypothesized in H4b, it was also found that discourse logic is the dominant factor increasing customer engagement in social for a product that is either at the growth or maturity stage. As the product matures, the positive tone of a social media post also increases customer engagement with a social media post. The current literature also supports this finding. Since customers are already familiar with the product and customer demand for the product is also increasing simultaneously, promotive messages trigger customers to engage with a product to a greater extent [103].

Results also confirmed our H4c, which indicates that product lifecycle positively moderates the correlation between argument frame and customer engagement in social media for a product at the decline stage. In this regard, as shown in Table 6, the positive argument frame of social media post for a product at the decline stage increases customer engagement in social media. At the decline stage, the effect of popularity was also found to increase customer engagement in social media.

Table 5
Discriminant Validity Analysis.

Fornell-Larcker Criterion				
	Customer engagement	Discourse logic	Argument frame	Popularity
Customer Engagement	1			
Discourse Logic	0.217	1		
Argument Frame	−0.089	−0.493	1	
Popularity	0.486	0.132	−0.023	1

Heterotrait-Monotrait Ratio (HTMT)			
	Discourse logic	Argument frame	Popularity
Discourse logic	1		
Argument frame	0.493	1	
Popularity	0.132	0.023	1

Cross loadings				
	Customer engagement	Discourse logic	Argument frame	Popularity
Popularity	0.486	0.132	−0.023	1
Discourse logic	0.217	1	−0.493	0.132
Argument frame	−0.089	−0.493	1	−0.023
Number of likes	0.763	0.169	−0.061	0.370
Number of shares	0.733	0.157	−0.074	0.357
Number of comments	0.005	0.013	−0.074	0.357

6. Post-hoc analysis – the effect of social media type on the antecedents of customer engagement in social media

To satisfy the study's fourth objective, we conducted a series of post hoc analyses. Results are shown in Figs. 4 and 5.

The analyses shown in the above figures revealed that the effect of social media influencer's popularity on customer engagement is similar between Instagram and Twitter. This finding was in line with our proposed hypothesis. On the other hand, our analysis results on discourse logic came back significantly different on our databases for Instagram and Twitter. The results showed that an *objectively* written social media post on Twitter is positively correlated with customer engagement. However, on Instagram, *subjectively* written social media posts were associated with customer engagement, contrary to our initial hypothesis. Although this finding was unexpected, it is in line with the User Gratification Theory [77]. According to this theory, individuals of different age groups have different tastes in using social media types and messages appealing to them. Therefore, due to users' demographic differences between the two platforms, social media users have different tastes for appealing messages, resulting in different customer engagement between Instagram and Twitter.

Like our previous finding, the social media post's argument frame results differed significantly between our two databases. Our analysis showed that on Instagram, users tend to engage higher with the posts that are written positively. On Twitter, social media users' engagement is not affected by the argument frame of a post. These two last findings and the previous section's findings reveal that firms need to employ different marketing strategies to attract users and increase their customer engagement in social media.

Finally, in line with our third and fourth research objectives, we checked to what extent product lifecycle moderates the relationships in this study's research model and to what extent these effects vary between Instagram and Twitter. The results are shown in Tables 7 and 8,

Table 6
Product lifecycle moderation effect on the hypothesized relationships.

Product lifecycle	R^2	Relationships	Moderation
Introduction stage	0.15	**Popularity → Customer engagement (0.253 ***)**	**Significant**
		Discourse logic → Customer engagement (0.228 *)	**Significant**
		Argument frame → Customer engagement (0.001 ns)	Not significant
Growth stage	0.16	Popularity → Customer engagement (0.049 ns)	Not significant
		Discourse logic → Customer engagement (0.399 **)	**Significant**
		Argument frame → Customer engagement (0.004 ns)	Not significant
Maturity stage	0.24	Popularity → Customer engagement (0.009 ns)	Not significant
		Discourse logic → Customer Engagement (0.427*)**	**Significant**
		Argument frame → Customer engagement (0.185 *)	**Significant**
Decline stage	0.11	**Popularity → Customer engagement (0.311 ***)**	**Significant**
		Discourse logic → Customer engagement 0.043 ns	Not significant
		Argument frame → Customer engagement (0.125 **)	**Significant**

ns, means not significant.
* $P < 0.05$.
** $P < 0.01$.
*** $P < 0.001$.

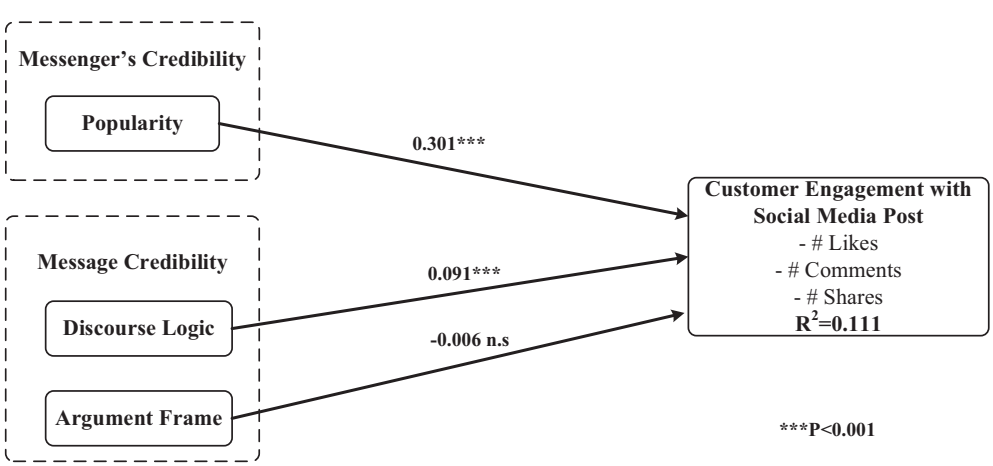

Fig. 3. Structural analysis results.

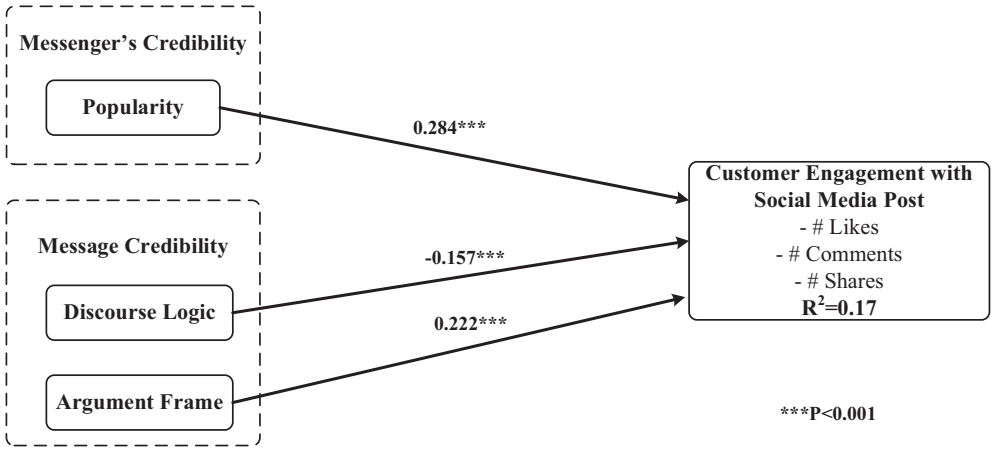

Fig. 4. Structural analysis results on Instagram.

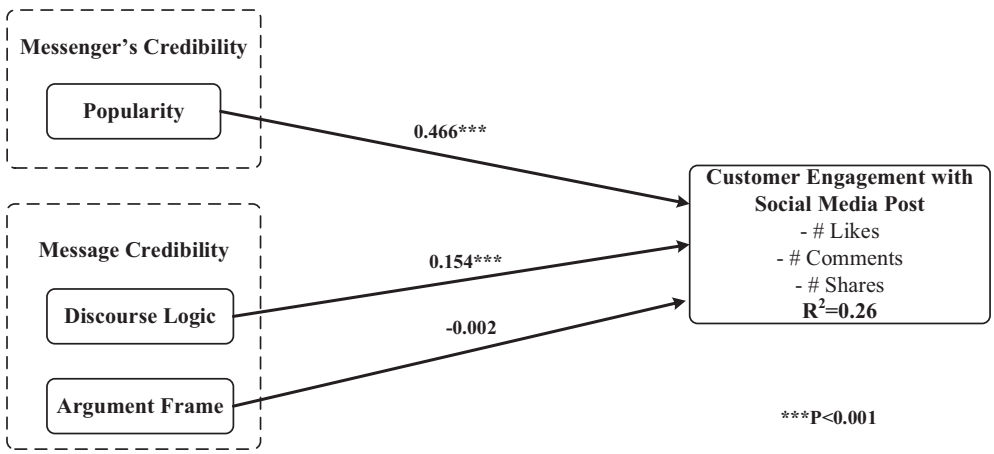

Fig. 5. Structural analysis results on Twitter.

respectfully, for Instagram and Twitter.

The above results confirm that across both platforms, as hypothesized in H4a, if a product is in its introduction stage, popularity's effect on customer engagement in social media is at its highest, positively increasing customer engagement in social media. Interestingly, it was also found that discourse logic at the introduction stage of a product only correlates with a higher level of customer engagement only on Twitter, such that the objectivity of social media posts on Twitter leads to a higher level of customer engagement in social media.

Our results on growth and maturity stages for the Twitter database supported our initial H4b hypothesis. In this regard, it was found that social media users on Twitter will engage to a greater extent with the objectively endorsed posts. Contrarily, it was found that on Instagram, subjectively written posts bring more customer engagement. It was also found that argument frame and popularity increase customer engagement in social media for products at their growth or maturity product lifecycle stages. These contradictory findings could mainly route the different demographic types of users between these two platforms since younger adults are more motivated by subjective opinions to participate in dialogue while older adults engage with objective views to a higher extent [104].

In line with the H4c hypothesis, our analysis showed that the positively written post (i.e., positive argument frame) increases customer engagement in social media for products at their decline stage. It was also found that popularity plays a role in driving customer engagement in social media for a product at its decline stage.

To sum, the results above analysis satisfied the third and fourth objectives of this study to understand the extent to which product lifecycle moderates the proposed antecedents of customer engagement in social media while showing that these effects vary by the social media platform.

7. Discussion

This study explored one probable explanation for customer engagement in social media through the lens of social media users' participation in the content of (co)creation. This study built on source credibility and heuristic/systematic processing of information theories to indicate how customer engagement in social media is shaped and affected by the factors related to a message (i.e., social media post) and its messenger's credibility. In this regard, it was found that customer engagement in social media is shaped by the number of *likes* and *shares* of a social media post. Surprisingly, it was also found that the number of comments does not significantly contribute to forming the concept of customer engagement in social media. Our findings showed that a product promoter's popularity in social media (i.e., a user whom many social media users follow) is a significant determinant of increasing customer engagement in social media across social media platforms. It was also found that, in general, objectively written social media posts increase customer engagement. Our initial analyses also revealed that a social media post's argument frame does not correlate with customer engagement on social media.

This study also examined the moderating effect of product lifecycle on the above correlations. Accordingly, at the introduction stage, it was

Table 7
The Moderating Effect of Product Lifecycle on Instagram Extracted Social Media Posts.

Product lifecycle	R^2	Relationships	Moderation
Introduction	0.60	**Popularity➔ Customer engagement ($P = 0.737$ ***)**	**Significant**
		Discourse Logic➔ Customer engagement ($P = -0.061$ ns)	Not significant
		Argument Frame➔ Customer engagement ($P = 0.09$ ns)	Not significant
Growth	0.23	**Popularity➔ Customer engagement ($P = 0.271$ *)**	**Significant**
		Discourse Logic➔ Customer engagement ($P = -0.278$ **)	**Significant**
		Argument Frame➔ Customer engagement ($P = 0.338$ **)	**Significant**
Maturity	0.20	**Popularity➔ Customer engagement ($P = 0.241$ ***)**	**Significant**
		Discourse logic➔ Customer engagement ($P = -0.147$ **)	**Significant**
		Argument Frame➔ Customer engagement ($P = 0.330$ *)**	**Significant**
Decline	0.31	**Popularity➔ Customer engagement ($P = 0.486$ ***)**	**Significant**
		Discourse logic➔ Customer engagement ($P = -0.025$ ns)	Not significant
		Argument frame➔ Customer engagement ($P = 0.200$ *)**	**Significant**

ns, means not significant.
* $P < 0.05$.
** $P < 0.01$.
*** $P < 0.001$.

Table 8
The moderating effect of product lifecycle on Twitter extracted social media posts.

Product lifecycle	R^2	Relationships	Moderation
Introduction	0.12	**Popularity➔ Customer engagement ($P = 0.159$ *)**	**Significant**
		Discourse logic➔ Customer engagement ($P = 0.327$*)**	**Significant**
		Argument frame➔ Customer engagement ($P = -0.003$ ns)	Not significant
Growth	0.61	Popularity➔ Customer engagement ($P = 0.182$ ns)	Not significant
		Discourse logic➔ Customer engagement ($P = 0.368$ **)	**Significant**
		Argument frame➔ Customer engagement ($P = 0.044$ ns)	Not significant
Maturity	0.16	Popularity➔ Customer engagement ($P = -0.208$ ns)	Not significant
		Discourse logic➔ Customer engagement ($P = 0.781$ *)**	**Significant**
		Argument frame➔ Customer engagement ($P = -0.001$ ns)	Not significant
Decline	0.16	**Popularity➔ Customer engagement ($P = 0.376$ ***)**	**Significant**
		Discourse logic➔ Customer engagement ($P = 0.039$ ns)	Not significant
		Argument frame➔ Customer engagement ($P = 0.087$ **)	**Significant**

ns, means not significant.
* $P < 0.05$.
** $P < 0.01$.
*** $P < 0.001$.

found that popularity and discourse logic increase engagement. As the product moves towards its growth and maturity stages, the objectivity of discourse logic provided on a post in social media becomes the primary driver of customer engagement in social media. Finally, it was found that for a product at the decline stage, the positivity of the argument frame and popularity of a promoter increase customer engagement in social media.

As discussed in our post hoc analysis, we also investigated to what extent the above findings vary by social media platforms (i.e., Instagram and Twitter). Our analyses revealed that popularity is one of the drivers of customer engagement in social media for both platforms. Interestingly, we found that the effect of discourse logic on customer engagement varies by the social media platform. While users on Instagram tend to participate in subjectively written posts, on Twitter, users engage with those posts objectively supported by influencers. Moreover, we found that while on Twitter, social media users tend to be neutral with the tone of a post, on Instagram, a positively written post invokes a higher level of customer engagement in social media. Finally, we explored the differences in product lifecycle's moderating effect on the above relationships between Instagram and Twitter. Our analyses showed that while popularity increases customer engagement in social media across both platforms at the introduction stage, results were contradictory for the growth and maturity stages. In general, our findings illustrated that on Twitter, social media posts' objectivity increases customer engagement for both growth and maturity stages. However, on Instagram, the popularity of a promoter and subjectivity and positivity of her/his post enhance customer engagement in social media. Finally, at the decline stage, results were similar across both platforms. At this stage, our analyses revealed that the popularity and positivity of a post's argument frame could improve customer engagement in social media.

7.1. Theoretical contributions

This study extends the extant literature on customer engagement by conceptualizing and validating the concept of customer engagement in social media through the lens of social media users' participation in content (co)creation. This concept can be used as a proxy to measure the current or potential future customers' engagements with a product service.

This study also extends the literature by introducing and validating popularity, discourse logic, and argument frame as the main factors that impact consumer engagement in social media. Specifically, we extended source credibility and HSM theory to the context of consumer engagement in social media. Our findings showed that a product promoter's popularity is a significant determinant of increasing customer engagement in social media across social media platforms.

Moreover, this study extends the literature by examining the role of product lifecycle on enhancing consumer engagement when using a heuristic cue (here argument frame, popularity) or systematic cue (here discourse logic). The findings showed that some of the above correlations vary by the stage a product is at its lifecycle. Our analyses show that popularity increases customer engagement with a social media post for a product at the introduction stage. For a product at the growth stage, we found that discourse logic (i.e., objectivity) is the only factor increasing customer engagement with a social media post. Finally, As the product matures, the argument frame's effect in enhancing customer engagement becomes more important, especially for products at their decline stage.

Finally, this study extends the literature by providing evidence that the effect of the factors affecting customer engagement in social media varies by the social media platform. Specifically, it was found that the antecedents of customer engagement in social media vary by platform. For instance, while on Instagram, subjectively and positively written social media posts bring more customer engagement for a product, on Twitter, social media users get engaged more with objectively and neutrally written social posts. This is a novel and interesting finding that shows the importance of social media type on understanding customer engagement. It moreover illustrates the need to study further the additional factors that impact customer engagement in different social media types, which could have a potential impact on firms' gaining competitive

advantage.

7.2. Practical contributions

This study also makes several practical contributions. As the first practical contribution, this study provided metrics to measure customer engagement in social media. Although on a practical level, indices such as Klout exist to show how social media users respond to post, no specific metrics can be found to measure and evaluate customer engagement with a promotive social media post. Having said that, this study argues that the number of likes and shares on a social media post as a good proxy to measure customer engagement with a social media post. This finding allows firms to measure how customers are responding to their customer engagement campaigns.

As another practical contribution, this study showed that renowned public figures' social media posts attract customers' attention and encourage them to participate in a social discourse regarding promoting a product on social media. This would assist firms to select influencers helpful in enhancing customer engagement properly.

As the third practical contribution, this study also showed that promotion and marketing strategies should vary by a product's lifecycle. Specifically, it was found that for a product at the introduction stage, popularity plays a significant role in enhancing customer engagement in social media. For newly introduced products application of this finding enables firms to enhance the attention of more social media users to their products. Moreover, for the products in growth or maturity phases, providing factual pieces of evidence enhances customer engagement. This finding implies that to increase customer engagement with a product, the promotion strategy should shift towards using objectively written social media posts. Moreover, for products at the decline stage, a positive argument frame increases customer engagement. This specific finding implies that providing positive promotive social media posts for a product at its decline phase can bring back social media users' attention to the product. These last two findings empower firms to avoid spending their time, money and effort on plans that do not bring value for them.

Finally, as another practical contribution, it was found that firms need to have different types of strategies for their social media posts across various platforms to enhance their customer engagement. These two findings help firms adjust their marketing strategies, plans and spending to increase customer engagement and, accordingly, sales and profit.

7.3. Limitation and future research

This study only focused on two primary social media platforms of Instagram and Twitter. Therefore, the generalizability of its findings across other social media platforms needs attention. Moreover, this study only selected posts regarding a high involvement product (i.e., vehicles). This study's results need to be validated by other research studies for different products or services types. Furthermore, in this study, we only used social media posts written in English for the sake of expertise. The classifiers used by this study should be adjusted for different cultures and languages to validate the results of the current study. Finally, in the present study, we only identified three antecedents for customer engagement in social media. Moreover, we just used secondary data for this study to validate the proposed hypotheses. Future research should find other customer engagement antecedents while employing other research methods to check the current study's findings' generalizability.

8. Conclusion

A significant number of unique individuals worldwide are using social media. Hence, many firms strive to utilize social media to enhance customer engagement, accordingly their sales, profit and customer loyalty. This study attempted to define, conceptualize and validate the concept of customer engagement in social media through the lens of customer (co)creation of content on social media. To that end, it was found that customer engagement in social media is shaped and can be measured by the number of likes and shares of a social media post. Moreover, the current study identified three main antecedents of customer engagement: popularity, discourse logic and argument frame. While focusing on social media posts related to four different categories of vehicles at different product lifecycle stages, this study used secondary data from two platforms of Instagram and Twitter to validate its research model and its associated hypotheses. Having said that, it was found that the effects of the identified antecedents of customer engagement in this study vary between the two chosen social media platforms. While on Instagram, social media users engaged to a greater extent with a positively written social media post that is subjectively supported by a promoter of a product, on Twitter, social media posts that are objectively supported bring more attention. It was also found that these effects vary by a product's lifecycle stage. Across both Instagram and Twitter and for a product in its introduction stage, popularity of a promoter plays an essential role in increasing customer engagement with a social media post. For products at their growth or maturity phases, in general, and across both Instagram and Twitter, discourse logic becomes the most imminent factor in increasing customer engagement with a social media post. Finally, products in their decline phase require more positively written posts to increase social media users' engagement.

References

[1] M. Dollarhide, Social Media Definition. https://www.investopedia.com/terms/s/social-media.asp, 2021. (Accessed 12 April 2021).

[2] S. Kemp, Digital 2021 - We Are Social. https://wearesocial.com/digital-2021, 2021. (Accessed 12 April 2021).

[3] H. Tankovska, Instagram - Statistics & Facts | Statista. https://www.statista.com/topics/1882/instagram/, 2021. (Accessed 12 April 2021).

[4] J. Park, H. Hyun, T. Thavisay, A study of antecedents and outcomes of social media WOM towards luxury brand purchase intention, J. Retail. Consum. Serv. 58 (2021) 102272, https://doi.org/10.1016/j.jretconser.2020.102272.

[5] Buffer, State of Social 2019 | Buffer. https://buffer.com/state-of-social-2019, 2019. (Accessed 12 April 2021).

[6] GlobalWebIndex, Top Global Consumer Trends in 2020 – GlobalWebIndex. https://www.globalwebindex.com/reports/trends-2020, 2020. (Accessed 12 April 2021).

[7] T. Powers, D. Advincula, M.S. Austin, S. Graiko, J. Snyder, Digital and social media in the purchase-decision process: a special report from the advertising research foundation, J. Advert. Res. 52 (2012) 479–489, https://doi.org/10.2501/JAR-52-4-479-489.

[8] K. Nelson-Field, Who should you impress (and where are they hiding)?, in: Atten. Econ. How Media Work Springer Singapore, 2020, pp. 123–138, https://doi.org/10.1007/978-981-15-1540-8_8.

[9] F. Simon, V. Tossan, Does brand-consumer social sharing matter? A relational framework of customer engagement to brand-hosted social media, J. Bus. Res. 85 (2018) 175–184, https://doi.org/10.1016/j.jbusres.2017.12.050.

[10] G.Y. Liao, T.C.E. Cheng, W.L. Shiau, C.I. Teng, Impact of online gamers' conscientiousness on team function engagement and loyalty, Decis. Support. Syst. 142 (2021) 113468, https://doi.org/10.1016/j.dss.2020.113468.

[11] C.M. Harmeling, J.W. Moffett, M.J. Arnold, B.D. Carlson, Toward a theory of customer engagement marketing, J. Acad. Mark. Sci. 45 (2017) 312–335, https://doi.org/10.1007/s11747-016-0509-2.

[12] F. de Oliveira Santini, W.J. Ladeira, D.C. Pinto, M.M. Herter, C.H. Sampaio, B. J. Babin, Customer engagement in social media: a framework and meta-analysis, J. Acad. Mark. Sci. 48 (2020) 1211–1228, https://doi.org/10.1007/s11747-020-00731-5.

[13] K.K.F. So, W. Wei, D. Martin, Understanding customer engagement and social media activities in tourism: a latent profile analysis and cross-validation, J. Bus. Res. 129 (2021) 474–483, https://doi.org/10.1016/J.JBUSRES.2020.05.054.

[14] E. Jaakkola, M.A.-J. of Service Research, The role of customer engagement behavior in value co-creation: a service system perspective, J. Serv. Res. 17 (2014) (2014) 247–261, https://doi.org/10.1177/1094670514529187.

[15] C. Cheung, M. Lee, X. Jin, Customer Engagement in an Online Social Platform: A Conceptual Model and Scale Development, ICIS 2011 Proc, in: https://aisel.aisnet.org/icis2011/proceedings/onlinecommunity/8, 2011. (Accessed 12 April 2021).

[16] K. Freberg, K. Graham, K. McGaughey, L.A. Freberg, Who are the social media influencers? A study of public perceptions of personality, Public Relat. Rev. 37 (2011) 90–92, https://doi.org/10.1016/j.pubrev.2010.11.001.

[17] K. Giffin, The contribution of studies of source credibility to a theory of interpersonal trust in the communication process, Psychol. Bull. 68 (1967) 104–120, https://doi.org/10.1037/h0024833.
[18] C. Ashley, T. Tuten, Creative strategies in social media marketing: an exploratory study of branded social content and consumer engagement, Psychol. Mark. 32 (2015) 15–27, https://doi.org/10.1002/mar.20761.
[19] InfluencerMarketingHub, What is an Influencer? - Social Media Influencers Defined [Updated 2021]. https://influencermarketinghub.com/what-is-an-influencer/, 2021. (Accessed 12 April 2021).
[20] E. Djafarova, C. Rushworth, Exploring the credibility of online celebrities' Instagram profiles in influencing the purchase decisions of young female users, Comput. Hum. Behav. 68 (2017) 1–7, https://doi.org/10.1016/j.chb.2016.11.009.
[21] I. Arapakis, M. Lalmas, B.B. Cambazoglu, M.-C. Marcos, J.M. Jose, User engagement in online news: under the scope of sentiment, interest, affect, and gaze, J. Assoc. Inf. Sci. Technol. 65 (2014) 1988–2005, https://doi.org/10.1002/asi.23096.
[22] S.P. Eslami, M. Ghasemaghaei, K. Hassanein, Which online reviews do consumers find most helpful? A multi-method investigation, Decis. Support. Syst. 113 (2018) 32–42, https://doi.org/10.1016/j.dss.2018.06.012.
[23] M. Salehan, D.J. Kim, Predicting the performance of online consumer reviews: a sentiment mining approach to big data analytics, Decis. Support. Syst. 81 (2016) 30–40, https://doi.org/10.1016/j.dss.2015.10.006.
[24] S. Klepper, Entry, exit, growth, and innovation over the product life cycle, Am. Econ. Rev. 86 (1996) 562–583, https://doi.org/10.2307/2118212.
[25] C.M. Armstrong, K. Niinimäki, S. Kujala, E. Karell, C. Lang, Sustainable product-service systems for clothing: exploring consumer perceptions of consumption alternatives in Finland, J. Clean. Prod. 97 (2015) 30–39, https://doi.org/10.1016/j.jclepro.2014.01.046.
[26] C. Forsey, Twitter, Facebook, or Instagram? Which Platform(s) You should Be On. https://blog.hubspot.com/marketing/twitter-vs-facebook, 2020. (Accessed 12 April 2021).
[27] C. Leone, Which Social Media Sites Get the Most Engagement?. https://www.webstrategiesinc.com/blog/which-social-media-sites-get-the-most-engagement, 2018. (Accessed 29 July 2021).
[28] S. Chen, S. Chaiken, The Heuristic-Systematic Model in its Broader Context. https://psycnet.apa.org/record/1999-02377-003, 1999. (Accessed 12 April 2021).
[29] C. Roberts, F. Alpert, Total customer engagement: designing and aligning key strategic elements to achieve growth, J. Prod. Brand. Manag. 19 (2010) 198–209, https://doi.org/10.1108/10610421011046175.
[30] B. Brown, J. Sikes, P. Willmott, Bullish On Digital: McKinsey Global Survey Results. http://kwu-alumni.org/moto/media/5558f00ce5b09.pdf, 2013. (Accessed 12 April 2021).
[31] V. Kumar, A. Pansari, Competitive advantage through engagement, J. Mark. Res. 53 (2016) 497–514, https://doi.org/10.1509/jmr.15.0044.
[32] S.D. Vivek, S.E. Beatty, R.M. Morgan, Customer engagement: exploring customer relationships beyond purchase, J. Mark. Theory Pract. 20 (2012) 122–146, https://doi.org/10.2753/MTP1069-6679200201.
[33] C.M. Sashi, G. Brynildsen, A. Bilgihan, Social media, customer engagement and advocacy: an empirical investigation using twitter data for quick service restaurants, Int. J. Contemp. Hosp. Manag. 31 (2019) 1247–1272, https://doi.org/10.1108/IJCHM-02-2018-0108.
[34] J. Carlson, M. Rahman, R. Voola, N. De Vries, Customer engagement behaviours in social media: capturing innovation opportunities, J. Serv. Mark. 32 (2018) 83–94, https://doi.org/10.1108/JSM-02-2017-0059.
[35] P. Harrigan, U. Evers, M. Miles, T. Daly, Customer engagement with tourism social media brands, Tour. Manag. 59 (2017) 597–609, https://doi.org/10.1016/j.tourman.2016.09.015.
[36] F. Cabiddu, M. De Carlo, G. Piccoli, Social media affordances: enabling customer engagement, Ann. Tour. Res. 48 (2014) 175–192, https://doi.org/10.1016/j.annals.2014.06.003.
[37] C. Prentice, X.Y. Han, L.L. Hua, L. Hu, The influence of identity-driven customer engagement on purchase intention, J. Retail. Consum. Serv. 47 (2019) 339–347, https://doi.org/10.1016/j.jretconser.2018.12.014.
[38] N.J. De Vries, J. Carlson, Examining the drivers and brand performance implications of customer engagement with brands in the social media environment, J. Brand Manag. 21 (2014) 495–515, https://doi.org/10.1057/bm.2014.18.
[39] A. Pansari, V. Kumar, Customer engagement: the construct, antecedents, and consequences, J. Acad. Mark. Sci. 45 (2017) 294–311, https://doi.org/10.1007/s11747-016-0485-6.
[40] V. Kumar, L. Aksoy, B. Donkers, R. Venkatesan, T. Wiesel, S. Tillmanns, Undervalued or overvalued customers: capturing total customer engagement value, J. Serv. Res. 13 (2010) 297–310, https://doi.org/10.1177/1094670510375602.
[41] E. Cucu, Instagram Engagement Stats in 2021 | Socialinsider. https://www.socialinsider.io/blog/instagram-engagement/, 2021. (Accessed 12 April 2021).
[42] C.L. Toma, Counting on Friends: Cues to Perceived Trustworthiness in Facebook Profiles. www.aaai.org, 2014. (Accessed 12 April 2021).
[43] J.C. McCroskey, T.A. Jenson, Image of mass media news sources, J. Broadcast. 19 (1975) 169–180, https://doi.org/10.1080/08838157509363777.
[44] N.R. LaPelle, R. Luckmann, E.H. Simpson, E.R. Martin, Identifying strategies to improve access to credible and relevant information for public health professionals: a qualitative study, BMC Public Health 6 (2006) 89, https://doi.org/10.1186/1471-2458-6-89.
[45] C. Edwards, P.R. Spence, C.J. Gentile, A. Edwards, A. Edwards, How much Klout do you have … a test of system generated cues on source credibility, Comput. Hum. Behav. 29 (2013) A12–A16, https://doi.org/10.1016/j.chb.2012.12.034.
[46] R. Kutthakaphan, W. Chokesamritpol, K.L. Examiner, L. Hallén, The Use of Celebrity Endorsement with the Help of Electronic Communication Channel (Instagram) Case Study of Magnum Ice Cream in Thailand COURSE CODE EFO705 AUTHORS the Use of Celebrity Endorsement with the Help of Electronic Communication Channel (Instagram): Case Study of Magnum Ice Cream Thailand. http://urn.kb.se/resolve?urn=urn:nbn:se:mdh:diva-19135, 2013. (Accessed 12 April 2021).
[47] S. Chaiken, A. Ledgerwood, A theory of heuristic and systematic information processing, in: Handb. Theor. Soc. Psychol. Collect Vol. 1 2, Google Books, 2011, in: https://books.google.ca/books?hl=en&lr=&id=0QuyCwAAQBAJ&oi=fnd&pg=PA246&dq=A+theory+of+heuristic+and+systematic+information+processing.+Handbook+of+theories+of+social+psychology:+Volume+&ots=-hAnOBl2gF&sig=Iq5UwipDOVCW_iNhKb7gZmPAc8o#v=onepage&q=Atheoryofheuristicandsystematicinformationprocessing.Handbookoftheoriesofsocialpsychology%3AVolume&f=false. (Accessed 12 April 2021).
[48] J. Cummings, A.R. Dennis, Virtual first impressions matter: the effect of enterprise social networking sites on impression formation in virtual teams, MIS Q. Manag. Inf. Syst. 42 (2018) 697–718, https://doi.org/10.25300/MISQ/2018/13202.
[49] M. Xiao, R. Wang, S. Chan-Olmsted, Factors affecting YouTube influencer marketing credibility: a heuristic-systematic model, J. Media Bus. Stud. 15 (2018) 188–213, https://doi.org/10.1080/16522354.2018.1501146.
[50] C.N. Wathen, J. Burkell, Believe it or not: factors influencing credibility on the Web, J. Am. Soc. Inf. Sci. Technol. 53 (2002) 134–144, https://doi.org/10.1002/asi.10016.
[51] A. Arora, S. Bansal, C. Kandpal, R. Aswani, Y. Dwivedi, Measuring social media influencer index- insights from Facebook, Twitter and Instagram, J. Retail. Consum. Serv. 49 (2019) 86–101, https://doi.org/10.1016/j.jretconser.2019.03.012.
[52] DigitalMarketingInstitute, 9 of the Biggest Social Media Influencers on Instagram | DMI. https://digitalmarketinginstitute.com/blog/9-of-the-biggest-social-media-influencers-on-instagram, 2018. (Accessed 12 April 2021).
[53] J. Cheriyan, G.P. Sajeev, SpreadMax: a scalable cascading model for influence maximization in social networks, in: 2018 Int. Conf. Adv. Comput. Commun. Informatics, ICACCI 2018, Institute of Electrical and Electronics Engineers Inc., 2018, pp. 1290–1296, https://doi.org/10.1109/ICACCI.2018.8554863.
[54] M.Y. Almoqbel, D.Y. Wohn, R.A. Hayes, M. Cha, Understanding Facebook news post comment reading and reacting behavior through political extremism and cultural orientation, Comput. Hum. Behav. 100 (2019) 118–126, https://doi.org/10.1016/j.chb.2019.06.006.
[55] I. Sen, S. Singh, A. Aggarwal, P. Kumaraguru, S. Mian, A. Datta, Worth its weight in likes: Towards detecting fake likes on instagram, in: WebSci 2018 - Proc. 10th ACM Conf. Web Sci., Association for Computing Machinery, Inc, New York, NY, USA, 2018, pp. 205–209, https://doi.org/10.1145/3201064.3201105.
[56] C. Prentice, X. Wang, X. Lin, An organic approach to customer engagement and loyalty, J. Comput. Inf. Syst. 60 (2020) 326–335, https://doi.org/10.1080/08874417.2018.1485528.
[57] L. Aksoy, A. van Riel, J. Kandampully, J. Wirtz, A. Den Ambtman, J. Bloemer, C. Horváth, B. Ramaseshan, J. Van de Klundert, Z. Gurhan Canli, Managing brands and customer engagement in online brand communities, J. Serv. Manag. 24 (2013) 223–244, https://doi.org/10.1108/09564231311326978.
[58] E. Uzunoğlu, S. Misci Kip, Brand communication through digital influencers: leveraging blogger engagement, Int. J. Inf. Manag. 34 (2014) 592–602, https://doi.org/10.1016/j.ijinfomgt.2014.04.007.
[59] C. Lou, S. Yuan, Influencer marketing: how message value and credibility affect consumer trust of branded content on social media, J. Interact. Advert. 19 (2019) 58–73, https://doi.org/10.1080/15252019.2018.1533501.
[60] T. Pucci, E. Casprini, C. Nosi, L. Zanni, Does social media usage affect online purchasing intention for wine? The moderating role of subjective and objective knowledge, Br. Food J. 121 (2019) 275–288, https://doi.org/10.1108/BFJ-06-2018-0400.
[61] M. Nisbet, Communicating climate change: why frames matter for public engagement, Environment. 51 (2009) 12–23, https://doi.org/10.3200/ENVT.51.2.12-23.
[62] P.C.S. Wu, Y.C. Wang, The influences of electronic word-of-mouth message appeal and message source credibility on brand attitude, Asia Pac. J. Mark. Logist. 23 (2011) 448–472, https://doi.org/10.1108/13555851111165020.
[63] K.M. Coleman, Fatal charades: Roman executions staged as mythological enactments, J. Roman. Stud. 80 (1990) 44–73, https://doi.org/10.2307/300280.
[64] J.P.B. Kosiba, H. Boateng, A.F. Okoe Amartey, R.O. Boakye, R. Hinson, Examining customer engagement and brand loyalty in retail banking: the trustworthiness influence, Int. J. Retail Distrib. Manag. 46 (2018) 764–779, https://doi.org/10.1108/IJRDM-08-2017-0163.
[65] M. Ozanne, S.Q. Liu, A.S. Mattila, Are attractive reviewers more persuasive? Examining the role of physical attractiveness in online reviews, J. Consum. Mark. 36 (2019) 728–739, https://doi.org/10.1108/JCM-02-2017-2096.
[66] K. Naumann, J. Bowden, M. Gabbott, Expanding customer engagement: the role of negative engagement, dual valences and contexts, Eur. J. Mark. 54 (2020) 1469–1499, https://doi.org/10.1108/EJM-07-2017-0464.
[67] T. Levitt, Exploit the Product Life Cycle, n.d. http://www.marketch.ru (accessed April 12, 2021).

[68] C.R. Anderson, C.P. Zeithaml, Stage of the product life cycle, business strategy, and business performance, Acad. Manag. J. 27 (1984) 5–24, https://doi.org/10.5465/255954.
[69] LearnMarketing, Promotion Through product Life Cycle - Business Frameworks-MKT 622. https://sites.google.com/site/frameworksmkt622/home/5-create-value-proposition/promotional-mix/promotion-through-product-life-cycle, 2020. (Accessed 12 April 2021).
[70] K. Lundin, 7 Proven Ways to Increase Customer Engagement on Social Media - Crowdspring Blog. https://www.crowdspring.com/blog/7-ways-to-increase-social-media-engagement/, 2018. (Accessed 29 July 2021).
[71] C.-T. Chiang, C.-F. Wei, K.R. Parker, B. Davey, Exploring the drivers of customer engagement behaviours in social network brand communities: towards a customer-learning model, J. Mark. Manag. 33 (2017) 1443–1464, https://doi.org/10.1080/0267257X.2017.1399922.
[72] H.C. Lin, P.F. Bruning, H. Swarna, Using online opinion leaders to promote the hedonic and utilitarian value of products and services, Bus. Horiz. 61 (2018) 431–442, https://doi.org/10.1016/j.bushor.2018.01.010.
[73] P.P. Saviotti, A. Pyka, The co-evolution of innovation, demand and growth, Econ. Innov. New Technol. 22 (2013) 461–482, https://doi.org/10.1080/10438599.2013.768492.
[74] Harvard Business Review, Exploit the Product Life Cycle. https://hbr.org/1965/11/exploit-the-product-life-cycle, 2020. (Accessed 12 April 2021).
[75] M. Lambkin, G.S. Day, Evolutionary processes in competitive markets: beyond the product life cycle, J. Mark. 53 (1989) 4–20, https://doi.org/10.1177/002224298905300304.
[76] M.T. Wroblewski, The Business Product Life Cycle of a Vehicle in Marketing. https://smallbusiness.chron.com/business-product-life-cycle-vehicle-marketing-21158.html, 2020. (Accessed 12 April 2021).
[77] T.E. Ruggiero, Uses and gratifications theory in the 21st century, Mass Commun. Soc. 3 (2000) 3–37, https://doi.org/10.1207/s15327825mcs0301_02.
[78] L. Bullock, Five Reasons why you Should Choose Instagram for your Next Influencer Marketing Campaign | Smart Insights. https://www.smartinsights.com/digital-marketing-strategy/five-reasons-why-you-should-choose-instagram-for-your-next-influencer-marketing-campaign/, 2018. (Accessed 12 April 2021).
[79] K. Harrison, Top Social Media Tips for every Platform Part 1: Facebook, Twitter And Instagram. https://www.forbes.com/sites/kateharrison/2018/07/02/top-social-media-tips-for-every-platform-part-1-facebook-twitter-and-instagram/?sh=28eaa3e91046, 2018. (Accessed 12 April 2021).
[80] K. Carmicheal, Social Media Impressions vs. Reach: What's More Important?. https://blog.hubspot.com/marketing/impressions-vs-reach, 2019. (Accessed 12 April 2021).
[81] R.A. Rather, L.D. Hollebeek, Customers' service-related engagement, experience, and behavioral intent: moderating role of age, J. Retail. Consum. Serv. 60 (2021) 102453, https://doi.org/10.1016/j.jretconser.2021.102453.
[82] N. Hajj, Y. Rizk, M. Awad, A subjectivity classification framework for sports articles using improved cortical algorithms, Neural Comput. & Applic. 31 (2019) 8069–8085, https://doi.org/10.1007/s00521-018-3549-3.
[83] G. Trigos, GitHub - Gustrigos/Subjectivity-Detector: Natural Language Processing Project Written in Python 3.6.5 to Determine Whether a Body of Text is Written Objectively or Subjectively. https://github.com/Gustrigos/Subjectivity-Detector, 2018. (Accessed 12 April 2021).
[84] S.J. Kim, E. Maslowska, A. Tamaddoni, The paradox of (dis)trust in sponsorship disclosure: the characteristics and effects of sponsored online consumer reviews, Decis. Support. Syst. 116 (2019) 114–124, https://doi.org/10.1016/j.dss.2018.10.014.
[85] B. Pang, L. Lee, A Sentimental Education: Sentiment Analysis Using Subjectivity Summarization Based on Minimum Cuts. http://arxiv.org/abs/cs/0409058, 2004. (Accessed 12 April 2021).
[86] R.M. Marcacini, R.G. Rossi, I.P. Matsuno, S.O. Rezende, Cross-domain aspect extraction for sentiment analysis: a transductive learning approach, Decis. Support. Syst. 114 (2018) 70–80, https://doi.org/10.1016/j.dss.2018.08.009.
[87] S.W.K. Chan, M.W.C. Chong, Sentiment analysis in financial texts, Decis. Support. Syst. 94 (2017) 53–64, https://doi.org/10.1016/j.dss.2016.10.006.
[88] Y. Liu, C. Jiang, H. Zhao, Assessing product competitive advantages from the perspective of customers by mining user-generated content on social media, Decis. Support. Syst. 123 (2019) 113079, https://doi.org/10.1016/j.dss.2019.113079.
[89] A. Collomb, D. Joyeux, O. Hasan, L. Brunie, A Study and Comparison of Sentiment Analysis Methods for Reputation Evaluation, 2014.
[90] B. Pang, L. Lee, Using Very Simple Statistics for Review Search: An Exploration, Manchester. www.eecs.umich.edu/, 2008. (Accessed 12 April 2021).
[91] J. Kazmaier, J.H. van Vuuren, A generic framework for sentiment analysis: leveraging opinion-bearing data to inform decision making, Decis. Support. Syst. 135 (2020) 113304, https://doi.org/10.1016/j.dss.2020.113304.
[92] A.L.D. Loureiro, V.L. Miguéis, L.F.M. da Silva, Exploring the use of deep neural networks for sales forecasting in fashion retail, Decis. Support. Syst. 114 (2018) 81–93, https://doi.org/10.1016/j.dss.2018.08.010.
[93] J.R. Landis, G.G. Koch, An application of hierarchical kappa-type statistics in the assessment of majority agreement among multiple observers, Biometrics. 33 (1977) 363, https://doi.org/10.2307/2529786.
[94] S.R.-J. of E.& M. Sciences, Research Design and Methods: A Systematic Review of Research Paradigms, Sampling Issues and Instruments Development. Pdfs.Semanticscholar.Org.
[95] J. Stigler, P. Gonzales, T. Kwanaka, S. Knoll, A. Serrano, Study: Methods and Findings from an Exploratory Research Project on Eighth-Grade Mathematics Instruction in Germany, Japan, and the United States. A Research …. https://eric.ed.gov/?id=ED431621, 1999. (Accessed 29 July 2021).
[96] J.F. Hair, M. Sarstedt, L.M. Matthews, C.M. Ringle, Identifying and treating unobserved heterogeneity with FIMIX-PLS: part I – method, Eur. Bus. Rev. 28 (2016) 63–76, https://doi.org/10.1108/EBR-09-2015-0094.
[97] C. Ringle, D. da Silva, D. Bido, Structural equation modeling with the SmartPLS by Christian Ringle, Dirceu da Silva, Diógenes Bido:: SSRN, Brazilian J. Mark. 13 (2) (2015). Retrieved October 19, 2015,, Available at SSRN: https://ssrn.com/abstract=2676422. (Accessed 12 April 2021).
[98] D. Gefen, D. Straub, A practical guide to factorial validity using PLS-GRAPH: tutorial and annotated example, Commun. Assoc. Inf. Syst. 16 (2005) 91–109. Doi:10.17705/1CAIS.01605.
[99] M. Wetzels, G. Odekerken-Schröder, C. Van Oppen, Using PLS path modeling for assessing hierarchical construct models: guidelines and empirical illustration, MIS Q. Manag. Inf. Syst. 33 (2009) 177–196, https://doi.org/10.2307/20650284.
[100] S. Cohen, Perceived Stress in a Probability Sample of the United States. - PsycNET, S. Spacapan S. Oskamp. https://psycnet.apa.org/record/1988-98838-002, 1988. (Accessed 13 April 2021).
[101] F. Falk, N. Miller, A Primer for Soft Modeling, Univ. Akron Press, 1992. https://psycnet.apa.org/record/1992-98610-000. (Accessed 12 April 2021).
[102] J.Q. Gronemus, P.S. Hair, K.B. Crawford, J.O. Nyalwidhe, K.M. Cunnion, N. K. Krishna, Potent inhibition of the classical pathway of complement by a novel C1q-binding peptide derived from the human astrovirus coat protein, Mol. Immunol. 48 (2010) 305–313, https://doi.org/10.1016/j.molimm.2010.07.012.
[103] AlexSoft, Product Life Cycle: How to Extend Product Market Presence | AltexSoft. https://www.altexsoft.com/blog/business/product-life-cycle-how-to-extend-your-product-market-presence/, 2019. (Accessed 12 April 2021).
[104] W.H. Weiger, M. Hammerschmidt, H.A. Wetzel, Don't you dare push me: how persuasive social media tactics shape customer engagement, J. Assoc. Consum. Res. 3 (2018) 364–439, https://doi.org/10.1086/698713.

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